

Toward Reliable Cuffless Blood Pressure Monitoring Using a Multimodal Ultrasound–ECG Wearable Platform

Syed Turab Haider Zaidi^{1,2}, Lee Youngjin¹, Shin Hur^{1,2}

¹Korea Institute of Machinery and Materials, Daejeon Metropolitan City, South Korea, ²University of Science and Technology, Daejeon Metropolitan City, South Korea

Background, Motivation and Objective

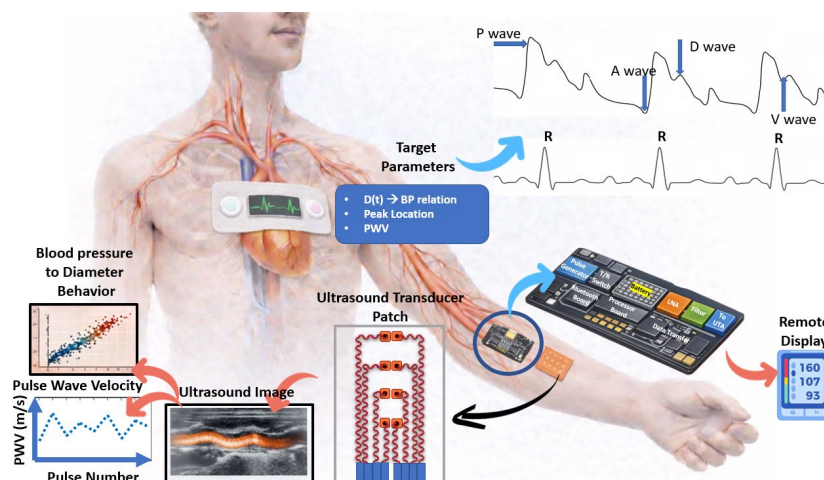
Hypertension remains a major global health challenge, affecting more than 1.3 billion people and contributing significantly to cardiovascular disease. Continuous blood pressure monitoring is essential for effective management, yet current approaches remain limited. Conventional arm cuffs provide only intermittent measurements and can disturb daily activities, while invasive arterial catheters are restricted to clinical environments. Existing cuffless wearable devices, often based on optical sensing or pulse transit time, suffer from motion artifacts and calibration drift. Recent studies suggest that improved reliability may be achieved by integrating complementary physiological signals, particularly electrical cardiac activity and mechanical vascular measurements.

Statement of Contribution/Methods

To address these limitations, we propose a flexible multimodal sensing platform integrating a PMN-PT 1-3 composite ultrasonic transducer array with a wearable ECG patch. The ultrasonic array is designed for conformal attachment over superficial arteries to monitor arterial wall motion and vessel diameter dynamics. The ECG patch simultaneously records cardiac electrical activity, providing precise timing information for synchronization with vascular mechanical signals. Combining ultrasound-based vascular measurements with ECG-derived cardiac timing enables multimodal hemodynamic analysis of arterial dynamics.

Results/Discussion

The system will be fabricated using a scalable PCB-based manufacturing method to support practical wearable implementation. In previous validation experiments using pulsatile vascular phantoms, the ultrasonic transducer array successfully tracked arterial diameter variations and reconstructed systolic and diastolic blood pressure with deviations of approximately 4 mmHg compared with reference pressure sensors. Maintaining errors within the clinically relevant ~5 mmHg range demonstrates that integrating ultrasound-derived mechanical measurements with ECG timing can improve the robustness of cuffless blood pressure estimation and supports the development of next-generation wearable cardiovascular monitoring systems.



Conceptual illustration of the multimodal cuffless blood pressure monitoring system